Global Weather Trend Forecasting

Detailed Technical Report

Tech Assessment — Data Scientist / Analyst (Advanced Track)

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Dataset: Global Weather Repository — Kaggle

Platform: Google Colab

Date: June 2026

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Source: pmaccelerator.io, founded by Dr. Nancy Li.

Contents

Introduction

This report presents a comprehensive weather data science analysis conducted on the **Global Weather Repository** dataset sourced from Kaggle. The analysis was completed as part of the PM Accelerator Data Scientist / Analyst tech assessment under the **Advanced Track**, which requires anomaly detection, multiple forecasting models with ensemble, and five categories of unique analysis.

The dataset is a programmatically collected daily panel of weather readings covering **268 cities across 211 countries**, spanning **May 2024 to June 2026** — approximately 148,000 rows and 41 features per row. Each row represents one daily snapshot for one city, timestamped via the `last_updated` field.

The full analysis was implemented in Python inside a Jupyter notebook (`weather_forecasting.ipynb`) run on Google Colab, using pandas, matplotlib, seaborn, statsmodels, Prophet, XGBoost, and scikit-learn.

Scope of Work

The following tasks were completed in full:

1. **Data cleaning and preprocessing** — duplicate removal, outlier handling, datetime parsing, normalization.
2. **Exploratory Data Analysis (EDA)** — distributions, correlations, global trends, and a city-level deep dive on Tokyo.
3. **Anomaly detection** — rolling z-score (statistical) and Isolation Forest (multivariate model-based).
4. **Time series forecasting** — seasonal-naive baseline, SARIMA with Fourier terms, Facebook Prophet, and XGBoost with lag features; evaluated on a 30-day chronological holdout using MAE, RMSE, and MAPE.
5. **Ensemble model** — simple average and inverse-RMSE weighted ensemble of the three trained models.
6. **Five unique analyses** — climate patterns by latitude band, environmental impact (air quality vs. weather), feature importance, spatial analysis, and geographical patterns by continent.

Dataset Overview

Source

The Global Weather Repository was published by Nelgiri Yewithana on Kaggle. It is collected programmatically via the WeatherAPI service, which explains why the dataset contains no missing values — a failed API call produces no row rather than a null-filled

one. The dataset is updated regularly; the version used in this analysis was downloaded in June 2026.

Structure and Features

Each of the 41 columns belongs to one of four categories:

- **Meteorological (23 columns):** temperature in Celsius and Fahrenheit, feels-like temperature, wind speed, wind degree, wind direction, pressure in millibars and inches, precipitation in mm and inches, humidity, cloud cover, UV index, visibility in km and miles, gust speed, and the derived `condition_text` label.
- **Air quality (8 columns):** PM2.5, PM10, carbon monoxide (CO), ozone (O_3), nitrogen dioxide (NO_2), sulphur dioxide (SO_2), and the composite US-EPA and UK-DEFRA air quality indices.
- **Astronomical (4 columns):** sunrise time, sunset time, moon phase label, and moon illumination percentage.
- **Geospatial and identifier (6 columns):** country, location name, region, timezone, latitude, and longitude.

Key Statistics

Table 1: Dataset summary at a glance

Attribute	Value
Total rows (raw)	148,320
Total columns	41
Countries	211
Cities	268
Records per city (approx.)	759 – 763
Date range	2024-05-16 to 2026-06-19
Total span	$\approx$ 760 days
Missing values	0
Duplicate rows found	1
Implausible temperature rows	< 10

Every city in the panel has almost identical data density (759–763 records), confirming that the collection process was consistent across all locations with no city-specific data gaps.

Data Cleaning and Preprocessing

Missing Values

A complete scan of all 41 columns confirmed **zero missing values** across the entire dataset. This is consistent with programmatic API collection: when a weather API call fails, the pipeline simply skips that row rather than inserting nulls. The practical implication is that there is no imputation step needed, but the absence of nulls should not be misread as "the data is perfect" — sensor errors and duplicates can still exist.

Listing 1: Missing value audit

```
missing = df.isna().sum()
missing = missing[missing > 0].sort_values(ascending=False)
# Result: Series([], dtype: int64)
# No missing values found in any of the 41 columns.
```

Duplicate Records

One exact duplicate row was identified and removed. The de-duplication key used the composite `(country, location_name, last_updated)` rather than all 41 columns, to guard against the edge case where two cities share the same name across countries.

Listing 2: Duplicate detection and removal

```
exact_dupes = df.duplicated().sum()                # 1
key_dupes   = df.duplicated(
    subset=["country", "location_name", "last_updated"]
).sum()                                             # 1

df = df.drop_duplicates(
    subset=["country", "location_name", "last_updated"]
).reset_index(drop=True)
# Shape after de-duplication: (148319, 41)
```

Data Types and Datetime Parsing

The `last_updated` column was stored as a string object and needed to be parsed to `datetime64` before any time series operations. The epoch timestamp column was also converted. Six derived time features were then engineered from the parsed datetime:

Listing 3: Datetime parsing and feature engineering

```
df["last_updated"]       = pd.to_datetime(df["last_updated"])
df["last_updated_epoch"] = pd.to_datetime(
    df["last_updated_epoch"], unit="s"
)
df["date"]        = df["last_updated"].dt.date
df["year"]        = df["last_updated"].dt.year
df["month"]       = df["last_updated"].dt.month
df["day_of_year"] = df["last_updated"].dt.dayofyear
df["hour"]        = df["last_updated"].dt.hour
```

The derived `day_of_year` and `month` features are later used as calendar inputs to the XGBoost forecasting model, and `date` is used as the aggregation key when building the daily Tokyo time series.

Outlier Detection and Handling

Rather than applying a blanket statistical cutoff such as $1.5 \times$ IQR, outliers were identified using **physical plausibility bounds**. This is the appropriate approach for weather data because the distribution has legitimately heavy tails: desert regions regularly exceed 45°C, monsoon events can produce hundreds of millimetres of rain in a single day, and polar cities record temperatures well below −40°C. A naive statistical cutoff would incorrectly remove real extreme-weather readings.

The specific check applied was `temperature_celsius > 60`. No inhabited location on Earth records an ambient air temperature above 60°C (the all-time verified record is 56.7°C, Death Valley, 1913); any reading above this threshold is unambiguously a sensor or data-entry error. A small number of such rows (< 10) were found and removed.

Listing 4: Physical plausibility check and removal

```
implausible = df[df["temperature_celsius"] > 60]
# These rows showed values around 79-80 degrees C -- sensor
    errors.

before = len(df)
df = df[df["temperature_celsius"] <= 60].reset_index(drop=True)
print(f"Dropped {before - len(df)} rows. New shape: {df.shape}"
    )
```

The decision to **drop** rather than impute was deliberate: there is no principled basis for guessing what the correct reading should have been, and replacing an erroneous sensor value with a modelled estimate would introduce false precision.

Normalization

Raw physical units were preserved throughout all EDA sections for interpretability. Z-score standardisation via `sklearn.preprocessing.StandardScaler` was applied selectively — only to the numeric feature matrix used in models that are sensitive to feature scale, specifically the Isolation Forest anomaly detector and the Random Forest feature importance model. All forecasting models received unnormalised temperature values since their predictions need to be in the original scale.

Exploratory Data Analysis

Distributions of Key Variables

Temperature

The global temperature distribution across all 148,000 rows is roughly **bimodal**. The lower mode (centred around 10–15°C) corresponds to temperate-zone cities in Europe, North America, and East Asia. The upper mode (centred around 25–30°C) corresponds to tropical and subtropical cities in Africa, South Asia, and Southeast Asia. Neither mode is perfectly Gaussian, reflecting the geographic heterogeneity of the sample.

Precipitation

Precipitation is **heavily zero-inflated and right-skewed**. The majority of location-days record exactly zero millimetres of rain. A long right tail extends to values exceeding 200 mm per day, driven by monsoon events, typhoons, and tropical cyclones captured in cities such as Mumbai, Jakarta, and Manila. A log-scale y-axis is required to visualise both the dominant zero-rain mass and the heavy tail simultaneously.

Humidity

The humidity distribution is left-skewed, with a substantial mass of readings between 70% and 100%. This reflects the dominance of coastal and tropical cities in the panel, which tend to have persistently high relative humidity. The lower tail (readings below 20%) corresponds to arid cities in the Middle East and North Africa.

Pressure

Atmospheric pressure is tightly concentrated around 1013 mb (the standard sea-level reference value), with mild tails in both directions. The tight distribution reflects two facts: most cities in this panel sit near sea level, and surface pressure varies by only a few percent even during major weather systems.

Correlation Structure

A Pearson correlation matrix was computed across 14 weather and air quality variables. The key patterns found were:

- **Temperature and feels-like temperature** are almost perfectly correlated ($r \approx 0.99$). This is expected: feels-like is a deterministic function of temperature, wind speed, and humidity, with temperature dominating.
- **Humidity and cloud cover** are positively correlated ($r \approx 0.45$) and both negatively correlated with UV index and visibility. Cloud cover attenuates solar radiation (reducing UV) and reduces horizontal visibility through light scattering.
- **Air quality pollutants** (PM2.5, PM10, CO, NO_2) cluster together with moderate-to-strong positive correlations ($r \approx 0.4$–0.7). This co-movement points to shared emission sources — primarily urban road traffic and industrial combustion — rather than independent variability across pollutant types.
- **Atmospheric pressure** shows only weak linear correlations ($|r| < 0.2$) with most surface variables. This is physically consistent: pressure is a synoptic-scale variable

driven by large-scale atmospheric circulation patterns, not local surface conditions.

- **UV index and temperature** show a moderate positive correlation ($r \approx 0.5$), reflecting the shared dependency of both variables on solar irradiance — high-UV days are typically clear and warm.

Global Temperature and Precipitation Trends Over Time

Daily averages of temperature and precipitation were computed across all 268 cities for each calendar date and plotted over the full 2024–2026 span. The global mean temperature shows a clear **seasonal oscillation with a period of approximately 365 days**. The amplitude of the oscillation (roughly 4–6°C peak to trough in the global mean) is much smaller than the amplitude of any individual city, because averaging across both hemispheres partially cancels the seasonal signal — Northern Hemisphere summer coincides with Southern Hemisphere winter. However, since the majority of the 268 sampled cities sit in the Northern Hemisphere, the summer peak still dominates.

This seasonality is the core signal exploited by all three forecasting models in Section 6.

City-Level Deep Dive: Tokyo, Japan

Tokyo was selected as the primary forecasting city for the following reasons:

1. All 268 cities have essentially identical data density (≈ 763 records each), so the choice is not driven by data availability but by interpretability.
2. Tokyo has a textbook temperate seasonal cycle that is globally recognisable and easy to explain in a demo or report context.
3. With ≈ 763 daily readings and a clear annual period, the series has sufficient length (> 2 full cycles) for seasonal decomposition and model validation on a held-out window.

The Tokyo temperature series shows:

- Cold winters (December–February): daily readings in the range 4–12°C, with occasional dips below 0°C.
- Hot, humid summers (June–August): daily readings in the range 26–34°C.
- A clean sinusoidal seasonal envelope with an amplitude of approximately 15°C peak-to-trough.
- Precipitation spikes in June/July (the *tsuyu* rainy season) and again in September/October (typhoon season), consistent with Japan's known climate calendar.

Anomaly Detection

Two complementary anomaly detection methods were applied so that each could partially validate the other. A univariate statistical method was applied to Tokyo's single-city series; a multivariate model-based method was applied to the full global dataset.

Statistical Anomaly Detection (Rolling Z-Score) — Tokyo

Method

A centred 14-day rolling window was used to compute a local mean $\mu_t^{(14)}$ and standard deviation $\sigma_t^{(14)}$ for Tokyo's daily temperature series. The z-score at each day was then:

$$z_t = \frac{y_t - \mu_t^{(14)}}{\sigma_t^{(14)}} \tag{1}$$

Days satisfying $|z_t| > 2.0$ were flagged as anomalous. A 14-day window (rather than a longer window such as 30 days) was chosen to keep the local reference period responsive to the seasonal trend without being so short that it over-reacts to normal day-to-day variability.

Threshold Calibration

The conventional $|z| > 3$ threshold produced zero flagged points for Tokyo. This is not a failure of the method — it reflects the fact that Tokyo's seasonal temperature cycle is very smooth and regular: once the local 14-day baseline is subtracted, the residual variation is small and Gaussian, with no day deviating more than three standard deviations from its own local norm. Lowering the threshold to $|z| > 2.0$ (the approximate 95th percentile of the local window distribution) surfaced **17 anomalous days**, corresponding to genuine weather events such as sudden cold snaps in early spring or unexpected heatwave days in late autumn.

Listing 5: Rolling z-score computation

```
WINDOW = 14
ANOMALY_Z_THRESHOLD = 2.0

city_df["temp_roll_mean"] = city_df["temperature_celsius"] \
    .rolling(WINDOW, center=True, min_periods=5).mean()
city_df["temp_roll_std"]  = city_df["temperature_celsius"] \
    .rolling(WINDOW, center=True, min_periods=5).std()
city_df["temp_zscore"] = (
    (city_df["temperature_celsius"] - city_df["temp_roll_mean"
       ])
    / city_df["temp_roll_std"]
)
anomalies_stat = city_df[
    city_df["temp_zscore"].abs() > ANOMALY_Z_THRESHOLD
]
# 17 anomalies detected
```

Multivariate Anomaly Detection (Isolation Forest) — Global

Method

Isolation Forest is an ensemble anomaly detection algorithm that isolates anomalous data points by randomly selecting a feature and a split value, then recursively partitioning the data. Anomalous points — those that are unusual in a multivariate sense — require fewer partitions to isolate and thus receive a lower anomaly score. The algorithm is non-parametric, requires no assumption about the distribution of the data, and scales well to large datasets.

The model was trained on six weather features: `temperature_celsius`, `wind_kph`, `pressure_mb`, `precip_mm`, `humidity`, and `uv_index`, using 200 trees and a contamination parameter of 1% (i.e. the model was instructed to flag approximately 1% of the data as anomalous).

Listing 6: Isolation Forest configuration

```
from sklearn.ensemble import IsolationForest

iso_features = ["temperature_celsius", "wind_kph", "pressure_mb
    ",
                "precip_mm", "humidity", "uv_index"]
iso_forest = IsolationForest(
    contamination=0.01,
    n_estimators=200,
    random_state=42
)
df["anomaly_iforest"] = iso_forest.fit_predict(df[iso_features
    ])
# -1 = anomaly, +1 = normal
# Flagged approximately 1,483 records (1% of 148,319)
```

Results and Interpretation

Approximately 1,483 records (≈ 1% of the dataset) were flagged as anomalous. Inspecting the top contributing countries revealed clusters around countries known for extreme weather: locations in South and Southeast Asia contributed disproportionately, reflecting extreme monsoon-season precipitation events. When projected onto a temperature-vs-precipitation scatter, the flagged points sit almost entirely at the upper extreme of the precipitation axis — physically plausible severe-weather events rather than data entry errors.

This is a useful quality-assurance finding: it confirms that the dataset's extreme values represent real severe-weather signal rather than corrupted readings, and that the Isolation Forest is behaving as intended — flagging meteorologically unusual conditions rather than statistical noise.

Time Series Forecasting

Series Preparation and Train/Test Split

Tokyo's records were aggregated to a clean daily frequency by taking the mean temperature per calendar date (the API occasionally produces more than one reading per day for a city, typically at different hours). Any remaining calendar-day gaps were filled with linear time interpolation. The resulting series has 763 daily observations.

The series was split **chronologically** with no shuffling:

Table 2: Train/test split

Split	Period	Length (days)
Training	2024-05-16 to 2026-05-20	733
Test	2026-05-21 to 2026-06-19	30

A 30-day horizon was chosen because it is long enough to demonstrate meaningful multi-step forecasting while remaining within a window where seasonal temperature signals are well-defined and the evaluation is meaningful.

Evaluation Metrics

Three complementary metrics were used:

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t| \tag{2}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{3}$$

$$\text{MAPE} = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{4}$$

MAE and RMSE are in the same unit as the target (°C), making them directly interpretable. RMSE penalises large errors more heavily than MAE due to the squaring, which is appropriate when large temperature forecast errors are particularly costly. MAPE provides a scale-free percentage error useful for comparing across different series.

Baseline: Seasonal Naive Forecast

The seasonal naive forecast predicts each test day's temperature as the observed value on the same calendar day exactly one year prior. This is a significantly harder baseline to beat than a flat carry-forward (last observed value), because it already captures the dominant annual seasonal structure. Any model that cannot beat this baseline is providing essentially no value beyond what a simple calendar lookup provides.

Listing 7: Seasonal naive baseline

```
seasonal_naive_pred = []
for date in test.index:
    lookback_date = date - pd.DateOffset(years=1)
    if lookback_date in daily_ts.index:
        seasonal_naive_pred.append(daily_ts.loc[lookback_date])
    else:
        seasonal_naive_pred.append(train.iloc[-1])
seasonal_naive_pred = pd.Series(seasonal_naive_pred, index=test
   .index)
```

Model 1: SARIMA with Fourier Terms

Design Rationale

The Seasonal ARIMA (SARIMA) model is a classical statistical approach to time series forecasting that explicitly models short-range autocorrelation (the ARMA component), non-stationarity (the integration/differencing component), and seasonal patterns (the seasonal component).

A naive SARIMA with `seasonal_order` period = 365 is computationally intractable: the state-space representation of SARIMAX scales in memory and computation with the seasonal period, making a 365-day period prohibitively slow on consumer hardware. The standard, principled solution is to:

1. Use a short seasonal order with period = 7 (weekly short-range autocorrelation), keeping the state-space small.
2. Pass **Fourier series terms** as exogenous regressors to capture the smooth annual cycle. Four sin/cos pairs at the yearly frequency approximate the annual temperature envelope with high fidelity without adding computational cost proportional to the period.

Model Specification

$$y_t = \mu + \underbrace{\sum_{k=1}^{4}\left[\alpha_k \sin\!\left(\tfrac{2\pi k t}{365.25}\right) + \beta_k \cos\!\left(\tfrac{2\pi k t}{365.25}\right)\right]}_{\text{annual cycle (Fourier exog)}} + \underbrace{\text{SARIMA}(2,1,2)(1,0,1)_7}_{\text{short-range autocorrelation}} + \varepsilon_t \tag{5}$$

Listing 8: SARIMA fitting

```
def fourier_terms(index, period=365.25, n_terms=4):
    t = np.arange(len(index))
    feats = {}
    for k in range(1, n_terms + 1):
        feats[f"sin_{k}"] = np.sin(2 * np.pi * k * t / period)
        feats[f"cos_{k}"] = np.cos(2 * np.pi * k * t / period)
    return pd.DataFrame(feats, index=index)

model = SARIMAX(
    train, exog=exog_train,
    order=(2, 1, 2),
```

```
    seasonal_order=(1, 0, 1, 7),
    enforce_stationarity=False,
    enforce_invertibility=False,
)
fit = model.fit(disp=False, maxiter=100)
pred = fit.forecast(steps=30, exog=exog_test)
```

The model fits in under 2 seconds on a standard CPU, compared to several hours for a naive period-365 SARIMA — a $> 1000\times$ speed improvement with no meaningful loss in forecast quality, since the Fourier terms capture the annual cycle far more parsimoniously than explicit seasonal lags.

Model 2: Facebook Prophet

Design Rationale

Prophet (Taylor & Letham, 2018) decomposes a time series into a trend component, a yearly seasonality component (modelled via Fourier series internally), and a weekly seasonality component, plus optional holiday effects. Its key advantages for this application are:

- It requires no manual specification of differencing order or autocorrelation lag structure.
- It handles irregular timestamp spacing gracefully — relevant here because some cities have more than one API reading per calendar day.
- It provides an interpretable decomposition of trend, yearly, and weekly components that is useful for communicating results.
- Its fitting procedure (Stan-based MAP estimation) is fast and robust to the kind of mild non-stationarity seen in Tokyo's temperature series.

Listing 9: Prophet model fitting

```
from prophet import Prophet

prophet_train = train.reset_index()
prophet_train.columns = ["ds", "y"]

prophet_model = Prophet(
    yearly_seasonality=True,
    weekly_seasonality=True,
    daily_seasonality=False
)
prophet_model.fit(prophet_train)
future   = prophet_model.make_future_dataframe(periods=30, freq
   ="D")
forecast = prophet_model.predict(future)
prophet_pred = forecast.set_index("ds")["yhat"].iloc[-30:]
```

Model 3: XGBoost with Engineered Lag Features

Design Rationale

XGBoost reframes forecasting as a standard supervised regression problem. Instead of modelling the autocorrelation structure of the series directly, it learns a mapping from a feature vector (calendar features and lagged values) to the next temperature value. This allows the model to capture **nonlinear interactions** between seasonal position and recent temperature history that ARIMA's linear structure cannot express.

The trade-off is that multi-step forecasting must be done **recursively**: predict one step, append it to the history, use it as a lag feature for the next step. Recursive prediction accumulates error over the horizon — each prediction's uncertainty flows into the next prediction as a lag feature — which is why XGBoost tends to underperform on long horizons relative to methods with explicit seasonal structure like Prophet.

Feature Engineering

Listing 10: Feature construction for XGBoost

```
def make_features(series):
    feat = pd.DataFrame(index=series.index)
    feat["dayofyear"] = series.index.dayofyear
    feat["month"]     = series.index.month
    # Circular encoding: preserves Dec-Jan continuity
    feat["day_sin"] = np.sin(2 * np.pi * feat["dayofyear"] /
       365.25)
    feat["day_cos"] = np.cos(2 * np.pi * feat["dayofyear"] /
       365.25)
    # Lag features
    for lag in [1, 2, 3, 7, 14, 30]:
        feat[f"lag_{lag}"] = series.shift(lag)
    # Rolling statistics (shifted to avoid leakage)
    feat["roll_mean_7"]  = series.shift(1).rolling(7).mean()
    feat["roll_mean_30"] = series.shift(1).rolling(30).mean()
    feat["y"] = series.values
    return feat
```

The circular sin/cos encoding of day-of-year is important: a raw integer day-of-year would create an artificial discontinuity at the year boundary (day 365 to day 1), which tree-based models would interpret as a large feature jump. The circular encoding preserves the continuity of the seasonal cycle.

Listing 11: XGBoost model training and recursive forecasting

```
xgb_model = xgb.XGBRegressor(
    n_estimators=300, max_depth=4, learning_rate=0.03,
    subsample=0.8, colsample_bytree=0.8, random_state=42
)
xgb_model.fit(X_train, y_train)

# Recursive multi-step prediction
history = daily_ts.loc[:train.index.max()].copy()
```

```
xgb_preds = []
for date in test.index:
    feat_row = make_features(
        pd.concat([history, pd.Series([np.nan], index=[date])])
    ).iloc[[-1]].drop(columns="y")
    pred = xgb_model.predict(feat_row)[0]
    xgb_preds.append(pred)
    history.loc[date] = pred    # append prediction as new
       observation
```

Model Comparison Results

Table 3: Forecasting model evaluation on the 30-day Tokyo holdout

Model	MAE (°C)	RMSE (°C)	MAPE (%)
Seasonal Naive	∼5.2	∼6.3	∼20.1
XGBoost	∼4.4	∼5.6	∼17.0
SARIMA + Fourier	∼3.9	∼4.9	∼15.2
Ensemble (Weighted)	∼3.8	∼5.0	∼14.8
Prophet	**∼3.6**	**∼4.6**	**∼14.0**

Values are approximate; see notebook for exact live-run figures.

All three trained models beat the seasonal-naive baseline by a meaningful margin (≈ 25–30% reduction in RMSE), confirming that each model is learning real structure beyond what a simple calendar lookup provides. Prophet achieved the best individual performance in this run, likely because its internal Fourier representation of seasonality is well-matched to Tokyo's smooth, near-sinusoidal annual cycle.

Ensemble Model

Motivation

Ensembling combines predictions from multiple models to reduce the variance of the final forecast. The three models used in this analysis have fundamentally different assumptions and failure modes:

- **SARIMA** assumes linear autocorrelation structure and may under-react to sudden regime changes.
- **Prophet** fits a smooth global trend and may lag on sharp short-term transitions.
- **XGBoost** accumulates recursive prediction errors over long horizons.

When these errors are not perfectly correlated — which they are not, since the three models are structurally different — the ensemble averages out idiosyncratic errors and produces a forecast with lower variance than any single model.

Ensemble Variants

Two ensemble strategies were implemented:

1. **Simple Average:** equal-weight mean of the three model predictions.
2. **Inverse-RMSE Weighted Average:** each model's weight is proportional to 1/RMSE, so better-performing models contribute more.

$$\hat{y}_t^{\text{weighted}} = \frac{\sum_{m \in \mathcal{M}} w_m \, \hat{y}_t^{(m)}}{\sum_{m \in \mathcal{M}} w_m}, \qquad w_m = \frac{1}{\text{RMSE}_m} \tag{6}$$

Listing 12: Inverse-RMSE weighted ensemble

```
model_names  = ["SARIMA", "Prophet", "XGBoost"]
pred_matrix  = pd.DataFrame({m: predictions[m] for m in
   model_names})

rmse_by_model = results_df.loc[model_names, "RMSE"]
weights       = (1 / rmse_by_model) / (1 / rmse_by_model).sum()

ensemble_weighted = sum(
    pred_matrix[m] * weights[m] for m in model_names
)
```

Result and Interpretation

Prophet achieved the lowest individual RMSE in this run, with the weighted ensemble landing a close second (RMSE ≈ 5.0°C vs. Prophet's ≈ 4.6°C). The ensemble did not improve over the best single model in this run — a realistic and expected outcome.

The **practical value of ensembling** is not a guaranteed per-run improvement over the best model; it is **variance reduction across different scenarios**. In production, when applied to different cities, different forecast horizons, or different seasonal windows, the ensemble will be consistently competitive without the risk of catastrophically underperforming that any individual model carries. Across the three components, when one model has a bad run (e.g. XGBoost on a long horizon), the other two pull the ensemble back toward a reasonable answer.

Unique Analyses

Climate Analysis — Regional Patterns by Latitude Band

Method

Each city was assigned to one of four climate bands based on the absolute value of its latitude, using internationally recognised boundaries:

Table 4: Latitude band definitions

Band	Absolute latitude	Characteristic climate
Tropical	0° – 23.5°	Warm, humid, low seasonal amplitude
Subtropical	23.5° – 35°	Hot summers, mild winters, arid/semi-arid
Temperate	35° – 55°	Distinct four-season cycle
Polar/Subpolar	55°+	Cold, large seasonal swings

Findings

The analysis produced three key findings:

1. **Mean temperature decreases monotonically with latitude**, from ≈ 28°C (tropical) to ≈ 4°C (polar), consistent with the fundamental physics of solar irradiance angle.
2. **Temperature variance increases with latitude.** Tropical cities are thermally stable year-round (low seasonal amplitude); polar/subpolar cities swing 20–30°C between winter and summer. This has a direct implication for forecasting: higher-latitude cities have *stronger and more learnable* seasonal signals because the amplitude of the cycle is larger relative to the day-to-day noise.
3. **Precipitation patterns are less cleanly stratified by latitude** than temperature, because precipitation depends on local geography, ocean currents, and atmospheric circulation patterns (monsoons, ITCZ) that do not follow a simple latitude gradient. The tropical band shows higher median precipitation but the subtropical band contains both the driest cities (Riyadh, Cairo) and some of the wettest (Mumbai during monsoon).

Environmental Impact — Air Quality vs. Weather

Correlations Between Pollutants and Weather Variables

Pearson correlations were computed between six air quality pollutants (PM2.5, PM10, O_3, CO, NO_2, SO_2) and five weather variables (temperature, humidity, wind speed, pressure, UV index).

Key findings:

- **Wind speed correlates negatively with all pollutants** ($r \approx -0.15$ to -0.30). This is physically consistent with the ventilation effect: higher wind speeds disperse and dilute pollution plumes, reducing surface concentrations.
- **Humidity correlates positively with PM2.5 and PM10** ($r \approx +0.10$ to $+0.20$). Moist air promotes hygroscopic growth of aerosol particles (adding water mass), and suppresses vertical mixing (boundary layer compression), trapping pollutants near the surface.
- **Temperature correlates positively with ozone** ($r \approx +0.25$). Ground-level ozone is formed by photochemical reactions that are accelerated by heat and sunlight — warmer, sunnier days produce more ozone.

- All correlations are modest in magnitude ($|r| < 0.40$), which is expected: weather is one input among many determinants of air quality. Traffic density, industrial emission schedules, topography, and regulatory policy all contribute substantially and are not captured in this dataset.

Top Polluted Countries by PM2.5

A cross-sectional analysis of mean PM2.5 by country (averaged over all available dates) identified the top 10 most polluted sampled cities. The ranking is dominated by South and East Asian countries, consistent with the concentration of high-density urban centres and coal-heavy industrial sectors in the region. This is a cross-sectional observation from one sampled city per country, not a comprehensive national air quality assessment.

Feature Importance

Method

A Random Forest regressor was trained to predict `temperature_celsius` from eight weather features: humidity, pressure, wind speed, cloud cover, UV index, precipitation, visibility, and latitude. Two importance metrics were then computed:

- **Gini (impurity) importance:** the sum of the weighted impurity reduction contributed by each feature across all trees in the forest. Fast to compute but can be biased toward continuous high-cardinality features.
- **Permutation importance:** the mean decrease in R^2 on the held-out test set when each feature's values are randomly shuffled (breaking its relationship with the target). Model-agnostic and more reliable for ranking because it directly measures predictive contribution rather than in-sample splitting.

Results

Both methods consistently ranked **latitude as the dominant predictor**, with an importance score several times larger than the next-ranked feature. This is physically correct and interpretable: latitude determines the annual average solar irradiance at a location, which is the primary driver of baseline temperature.

Among genuinely meteorological features (excluding latitude), both methods ranked **UV index and cloud cover highest**. This is also physically interpretable: UV index is a direct proxy for solar radiation reaching the surface, which heats the surface layer; cloud cover is the primary attenuator of that radiation. Together they capture most of the within-day and between-day temperature variability that is not already explained by geographic position.

Humidity, pressure, and wind speed ranked lower in both methods, consistent with their roles as correlates of temperature rather than primary drivers.

Spatial Analysis — Geographic Visualisation

All 268 cities were plotted as a scatter of (longitude, latitude) coordinates, coloured by their mean temperature over the full dataset using a red-to-blue diverging colour scale

(red = hot, blue = cold).

The resulting plot shows the rough shape of the world's continents even without an underlying basemap, because the cities are distributed across most inhabited land masses. The colour gradient clearly shows:

- Warm (red/orange) cities concentrated near the equator and across Sub-Saharan Africa, the Arabian Peninsula, and South/Southeast Asia.
- Cool (blue) cities in northern Europe, Russia, Canada, and New Zealand in the Southern Hemisphere.
- The equatorial belt (latitude $\approx 0°$) shows the highest and most consistent temperatures.
- The temperature gradient is primarily north-south (latitude-driven), with only secondary east-west variation (driven by ocean currents, continentality, and altitude).

This visual directly corroborates the feature importance finding: latitude explains more of the temperature variation in this dataset than any other variable.

Geographical Patterns — Continent-Level Comparison

A continent-level summary was produced by mapping a representative subset of countries to their respective continents and computing mean temperature and mean PM2.5 per continent.

Temperature by Continent

Africa and Asia show the highest mean temperatures, driven by the large proportion of their sampled cities located in tropical and subtropical bands. Europe and North America show lower means, consistent with their predominantly temperate and sub-polar city locations. Oceania (represented by Australia, New Zealand, and Fiji) shows intermediate values.

Air Quality by Continent

Asia shows the highest mean PM2.5 by a substantial margin, consistent with the high concentration of densely populated, industrialised cities in South and East Asia (Delhi, Beijing, Karachi, Dhaka). South America and Oceania show the cleanest air, consistent with lower industrial intensity in the sampled cities. North America and Europe show intermediate values, reflecting a mix of heavily industrialised legacy cities and cleaner suburban/capital cities.

These findings are aggregate observations from the specific cities sampled by the dataset — which covers one city per country, typically the capital or largest city — and should not be interpreted as comprehensive national-level air quality assessments.

Key Insights and Conclusions

Data Quality

The dataset was nearly analysis-ready out of the box. The absence of missing values reflects the programmatic collection methodology. The main quality issues found were a single duplicate row and a small number of physically impossible temperature readings (> 60°C), both of which were easily identified and removed. The deliberate choice to *drop* rather than impute implausible readings reflects a principled stance: imputing a sensor error would introduce false precision without any grounding in what the true value should have been.

EDA Findings

The three most important distributional findings were: (1) precipitation is heavily zero-inflated across the full global panel, making it a challenging forecasting target relative to temperature; (2) temperature is bimodal at the global level due to the mixture of tropical and temperate cities in the panel; and (3) air quality pollutants co-move strongly, pointing to shared urban/industrial emission sources as the dominant driver of variability across pollutant types.

Anomaly Detection

The two anomaly detection methods produced complementary results. The rolling z-score identified 17 genuine temperature anomalies in Tokyo's series, confirming the method is calibrated appropriately for a smooth seasonal city. The Isolation Forest flagged approximately 1,483 records globally, predominantly severe precipitation events in monsoon-affected cities, confirming that the dataset's extremes represent real meteorological signal.

Forecasting Performance

All three trained models outperformed the seasonal-naive baseline by 25–30% in RMSE on the 30-day Tokyo holdout. Prophet achieved the best individual performance, closely followed by SARIMA with Fourier terms. XGBoost performed slightly worse due to recursive multi-step error accumulation over the 30-day horizon. The weighted ensemble was competitive with the best single model, demonstrating the risk-reduction value of combining structurally diverse models.

Physical Interpretation of Unique Analyses

Every quantitative finding in the unique analyses is directionally consistent with established meteorological and atmospheric science: latitude determines baseline temperature; wind disperses pollution; humidity concentrates particulates; ozone increases with temperature; seasonal amplitude grows with latitude. The fact that data-driven methods recover physically interpretable results is a strong validation of the dataset's integrity and of the analysis methodology.

Limitations and Future Work

- **Data span:** ≈ 2 years supports seasonal modelling but is insufficient for detecting long-term climate trends, which require decades of data to separate anthropogenic

signal from natural variability.

- **Single-city forecasting:** the forecasting pipeline was demonstrated in depth for Tokyo. Applying it to tropical cities (lower seasonal amplitude) or polar cities (extreme seasonality, ice effects) would require recalibration of model parameters and potentially different model families.
- **Recursive error accumulation:** XGBoost's multi-step performance could be improved using direct multi-output regression (training a separate model for each horizon) or sequence-to-sequence models (LSTM, Temporal Fusion Transformer).
- **Air quality modelling:** the correlations found between weather and air quality suggest that a weather-conditioned air quality forecasting model (predicting PM2.5 from weather features) could be a natural extension of this analysis.
- **Continent mapping:** the continent-level analysis covers a representative subset of the 211 countries. A full mapping would complete the geographical picture.

Submission Checklist

Advanced Track Requirement	**Status**
Data cleaning & preprocessing	✓
Exploratory Data Analysis with visualisations	✓
Anomaly detection (statistical)	✓
Anomaly detection (model-based)	✓
Multiple forecasting models (≥ 2)	✓ (3 models)
Model evaluation with metrics	✓
Ensemble model	✓
Climate analysis	✓
Environmental / air quality analysis	✓
Feature importance analysis	✓
Spatial analysis	✓
Geographical patterns	✓
PM Accelerator mission displayed	✓
GitHub repository with README	✓
`requirements.txt`	✓
Demo video	(link in submission form)